

Find A Gene Project Part 1

[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function.

Protein: Kinesin Family Member 11 (KIF11)

Species: Homo sapiens

Accession Number: NM_004523.4 (RefSeq protein)

Function: KIF11 is a kinesin motor protein that helps separate chromosomes during mitosis by organizing and maintaining the bipolar spindle. It uses ATP-powered movement along microtubules to support proper cell division.

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied...

A BLASTN search was performed using the nucleotide sequence of the human KIF11 gene (accession: NM_004523.4) against the Transcriptome Shotgun Assembly (TSA) database limited to bacteria (taxid:2).

Organism: Bacteria

“Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i5”
(Accession: GFOJ010595215.1)

- **Max Score: 344**
- **E-value: 2e-90**
- **Percent Identity: 68.06%**
- **Query Coverage: 21%**

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🔔 Your search is limited to records include: Bacteria (taxid:2)

Job Title

NM_004523:Homo sapiens kinesin family member...

RID

0TBN955D014

Search expires on 05-21 02:40 am

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Database

tsa (25 databases)

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Query ID

NM_004523.4

Description

Homo sapiens kinesin family member 11 (KIF11), mRNA

Molecule type

nucleic acid

Query Length

5016

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MSA viewer

🔔

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to

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to

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GenBank

Graphics

Distance tree of results

MSA Viewer

	Description	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☐	Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i5	344	344	21%	2e-90	68.06%	3425	GFOJ010595215.1
☐	Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i4	344	344	21%	2e-90	68.06%	2358	GFOJ010595214.1
☐	Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i6	344	344	21%	2e-90	68.06%	3303	GFOJ010559821.1
☐	Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i3	344	344	21%	2e-90	68.06%	2300	GFOJ010559812.1
☐	TSA: Rhodococcus rhodochrous TRINITY_DN1599_c4_g1_i1 transcribed RNA sequence	262	340	6%	6e-66	80.78%	317	GGWF01000084.1

Visual pairwise alignment:

TSA: Candidatus Thiodiazotropha endolucinida Codakia_V2_299248_c0_g4_i5, transcribed RNA sequence

Sequence ID: GFOJ010595215.1 Length: 3425 Number of Matches: 1

Range 1: 2106 to 3142 GenBank Graphics ▾ Next Match ▲ Previous Match

Score	Expect	Identities	Gaps	Strand
344 bits(381)	2e-90	714/1049(68%)	23/1049(2%)	Plus/Minus
Query 495	AAGTGTGTTGTTTGTCCAATTCTGGATGAAGTTATTATGGGCTATAATTGCACTATCTTTGC	554		
Sbjct 3142	AAGTATTGTGTACCAATAATTAGTGAAGTCTGAGTGGTTACAACGTACAATTTTTC	3083		
Query 555	GTATGGCCAACTGGCACTGGAAAACTTTTACAATGGAAGGTGAAAGGTACACCTAATGA	614		
Sbjct 3082	CTATGGCCAAACAGGGACCGGAAAAACCTTCACTATGGAAGGGGAGCGATCAG---ATGA	3026		
Query 615	AGAGTAT---ACCTGGGAAGAGGATCCCTTGGCTGGTATAATTCACGTACCCTTCATCA	671		
Sbjct 3025	CCCCTGTGTACCTGGGAAGATGACCCCTTACTGGTATCATCCCAAGAGCGCTGGCCAA	2966		
Query 672	AATTTTTGAGAACTTACTGATAATGGTACTGAATTTTCAGTCAAAGTGTCTCTGTTGGA	731		
Sbjct 2965	CATATTGTATGTTCTCACCACAGGAGGTGGAATTCTCCGTCGAGTGTCTTTTCTGGA	2906		
Query 732	GATCTAATGAAGAGCTTTTGTATCTTCTTAATCCATCATCTGATGTTTCTGAGAGACT	791		
Sbjct 2905	ACTCTACAATGAAGAGCTGTTTGATCTGCTTGGTTCATCAGAAGATCCA-CTGA-AGTTG	2848		
Query 792	ACAGATGTTTGATGATCCCGTAACAAGAGAGG-----AGTGATAATTAAAGGTTTAGAAG	847		
Sbjct 2847	AGA-ATATACGAAGA---CAGTG-CAAGGAAGGGCTCAGTAATAATCCATGGCCTAGAAG	2793		
Query 848	AAATTACAGTACACAACAAGGATGAAGTCTATCAAATTTTAGAAAAGGGGGCAGCAAAAA	907		

The selected hit has an E-value of 2e-90, a score of 344 bits, and 68% identity across the aligned region.

[Q3] Gather information about this “novel” protein. At a minimum, show me the protein sequence of the “novel” protein as displayed in your BLAST results from [Q2] as FASTA format...

Selected novel protein hit: Candidatus Thiodiazotropha endolucinida
Codakia_V2_299248_c0_g4_i5

Species: Candidatus Thiodiazotropha endolucinida

FASTA Sequence (translated ORF using EMBOSS Transeq):

```
>GFOJ010595215.1_5 TSA: Candidatus Thiodiazotropha endolucinida  
Codakia_V2_299248_c0_g4_i5  
EIFLQKI*EGLKQNLKSMPSQQKKEKSDQHIQVAVRLRPINAVEKKQCSYSVVDGSADRR  
EVTVKERLGINPTTKTFNFDHVFPPPEVKQVQIYKSIVSPIISEVLSGYNCTIFAYGQTGT  
GKTFTMEGERSDDPCVTWEDDPLTGII PRALANIFDVLTNQEVFEFSVRVSFLELYNEELF  
DLLGSSDPLKLRIYEDSARKGSVIIHGLEEMVVRSKDEVYDILERGAARRQTAATLLNA  
HSSRSHSVFTVTIHIKENTMEGEELLKTGKLHLVDLAGSENIGRSGAVDKRAREAGNINQ  
SLLTLGRVITALVEHAPHIPYRESKLTRLQDSLGGRTKTSIIATVSPASINLDETSTL  
DYAHRAKNITNRPEINQKLTKRALIKEYTEEIEKLRRDLQASREKNGFYIAEDNFNSMQM  
KLTVQQENIEEMTEKIEVLSDDELTKTQELFSDTKKDLEQTTEKLTETTENLRNTKVTLKE  
TKITLRDTEQDRNEQRFLVDEHVKTETDLYSQATELLGTVGESTNDVRLHEKVDRKAV  
ESHNEECQGSFQRKYHQDIQSMMDELKGFLNEQETSHNAHTEHIAQKVAKNKHVSELTS  
EVIELQKGLQERVESMNSQQKTCACNSMSWMKDLKNSYLTKEEEIQSIGKFEEEMISKS  
TAVVSKLQNMTESLDQLSSLIEQQKQEEEEQRLQNWADRLTGDLDKLNSLVTQYSSRQLET  
MTGLKTHVVDTQTEYNATKQEI DVKMNEIFQIADKLKELNTLLAKNSSTVTQHSNHTLSE  
IEKSSSSAEDMSSNLNSEINHIKERSNSCLQESQEKLESNCQCVHKDIDKIKSVNTTTVD  
IQTEIQTYNTVGKQQRIKFLDSYQDTLEDRCTEHETTINTDLNVFYKNTDALQSENQSAC  
DKLISSLDSTASFDEAVATENQLLISRISGVRNFSQDQSARLTDRSDVQRFVGEELRK  
DIPTGQTPVRKDFQYPRDLKKTGRHSLLEKFREDNMVNLSLDEDSIVEESDEIKETEMQ  
NDDVADGKG*CC* KLLNIRCIFSISPO*G* GYRISGK*KTRSTPGI*KKSEKKFQKCY  
TKETDQTATENSYAKWTVKLNFRVNLLQIKQ*RKRLWWIFGDNWII FSSSP*NYIF*VL  
IX
```

[Q4] Prove that this gene, and its corresponding protein, are novel...

A BLASTP search of the translated protein sequence against the nr database showed that the top match (kinesin-like protein KIF11-B [Mya arenaria]) had 54.66% amino acid identity to a protein from a different species, Mya arenaria. To conclude: this protein is likely novel because there is no 100% identical match from the same species.

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Job Title

GFOJ010595215.1_5 TSA: Candidatus Thiodiazotropha...

RID

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Database

ClusteredNR [See details ▾](#)

Query ID

lcl|Query_1034714

Description

GFOJ010595215.1_5 TSA: Candidatus Thiodiazotropha en...

Molecule type

amino acid

Query Length

1142

Other reports

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to

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to

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	Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	1 member(s), 1 organism(s)	softshell	kinesin-like protein KIF11-B [Mya arenaria]	1038	1038	87%	0.0	54.66%	1094	XP_052770351.1
☒	1 member(s), 1 organism(s)	bivalves	unnamed protein product [Chamelelea gallina]	1031	1031	87%	0.0	53.14%	1077	CAN1469806.1
☒	1 member(s), 1 organism(s)	northern quahog	kinesin-like protein KIF11-B [Mercenaria mercenaria]	1026	1026	88%	0.0	52.72%	1077	XP_053406875.1
☒	1 member(s), 1 organism(s)	Manila clam	kinesin-like protein KIF11 [Ruditapes philippinarum]	994	994	88%	0.0	51.89%	1090	XP_060567348.1
☒	6 member(s), 1 organism(s)	zebra mussel	hypothetical protein DPMN_163994, partial [Dreissena polymor...	984	984	87%	0.0	51.98%	1129	KAH3785899.1
☒	2 member(s), 2 organism(s)	bivalves	hvothetical protein ACJMK2_005000 [Sinanodonta woodiana]	983	983	86%	0.0	52.65%	1086	KAL3863235.1

Feedback

Visual pairwise alignment:

kinesin-like protein KIF11-B [Mya arenaria]

Sequence ID: [XP_052770351.1](#) Length: 1094 Number of Matches: 1

Range 1: 1 to 985 [GenPept](#) [Graphics](#)

[Next Match](#) [Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
1038 bits(2684)	0.0	Compositional matrix adjust.	551/1008(55%)	714/1008(70%)	39/1008(3%)
Query 18	MPSQKKEKSDQHIQVAVRLRPINAVEKKQCSYSVVDGSDRREVTVKERLGINPTTKTF	77			
Sbjct 1	MP + ++ +OHIQVAVRLRPIN+ E+KQ SYSVVDG+A+++EVTVKERLG+NPTTKTF	60			
Query 78	NFDHVFPEVKQVQIYKSIVSPIISEVLSGYNCTIFAYGQTGTGKFTMEGERSDDPCVT	137			
Sbjct 61	NFDHVF +VKQ Q+YKSIVSPII EVL GYNCTIFAYGQTGTGKFTMEGERSDDP ++	120			
Query 138	NFDHVFQDVKQVQYKSIVSPIIDEVLDGYNCTIFAYGQTGTGKFTMEGERSDDPNMS				
Query 138	WEDDPLTGIIPRALANIFDVLTNQEVFVSVRVSFLELYNEELFDLLGSSDPLKRIYED	197			
Sbjct 121	WEDDPLTG+IPRALANIFD L Q VEFVSVRVSFLELYNEELFDLLGSS DPL+L+IYED	180			
Query 198	WEDDPLTGVIPRALANIFDQLQAQAVEFVSVRVSFLELYNEELFDLLGSSIDPLRLKIYED				
Query 198	SARKGSVIIHGLEEMVVRSKDEVYDILERGAARRQTAATLLNAHSSRSHSVFTVTIHIKE	257			
Sbjct 181	S RKGSVII GLEE+VV+SKDEVY+ILERGA+RRQTAATLLNA SSRHSVFTVTIH KE	240			
Query 258	STRKGSVIIQGLEEIVVKSDEVYIILERGASRRQTAATLLNAFSSRSHSVFTVTIHTKE				
Query 258	NTMEGEELLKTGKLHLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVEHAP	317			
Sbjct 241	NTMEGEELLKTGKLHLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVEHAP	300			
Query 318	NTMEGEELLKTGKLHLVDLAGSENIGRSGAVDKRAREAGNINQSLLTLGRVITALVEHAP				
Query 318	HIPYRESKLTRLLQDSLGGRTKTSIIATVSPASINLDETSTLDYAHRAKNITNRPEINQ	377			
Sbjct 301	H+PYRESKLTR+LQDSLGGRTKTSIIAT+SPAS+NL+ETLSTLDYAHRAKNITNRPE+NQ	360			
Sbjct 301	HVPYRESKLTRILQDSLGGRTKTSIIATISPASVNLEETLSTLDYAHRAKNITNRPEVNQ				

The selected hit has an E-value of 0, a score of 1038 bits, and 55% identity across the aligned region.